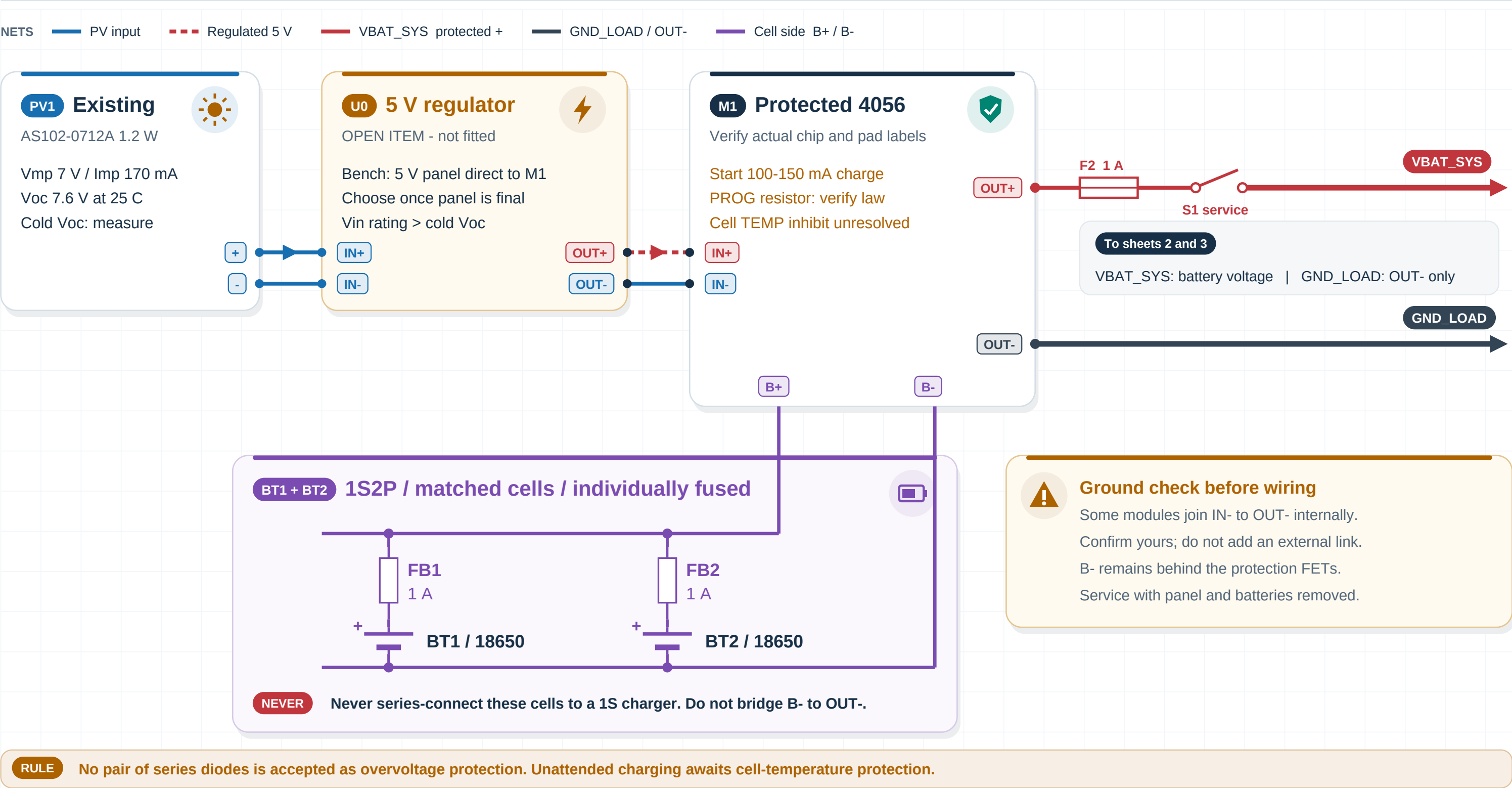


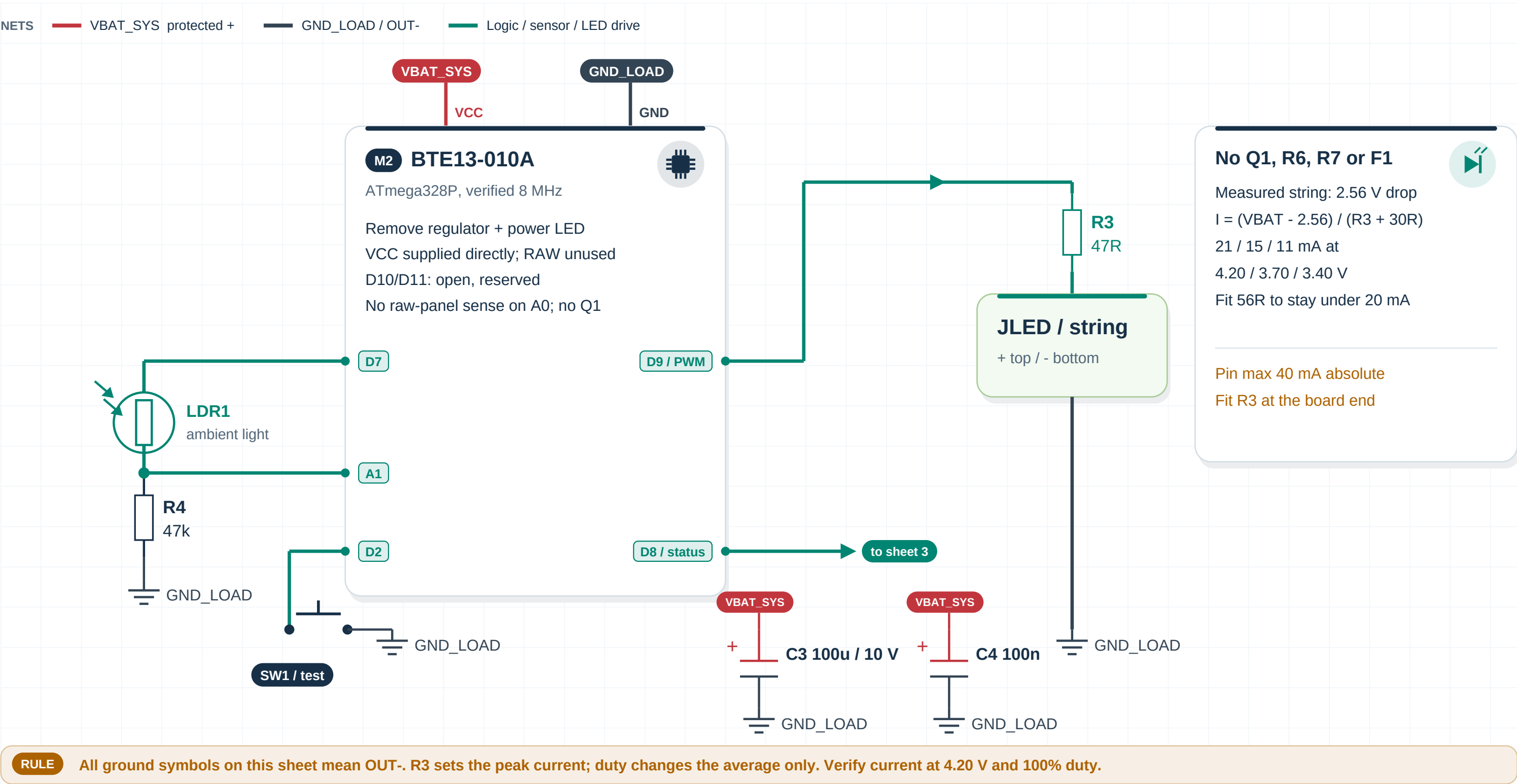
Power, charging & battery

Terminal-level schematic. Amber blocks are open items or need bench qualification. All loads connect to protected outputs.



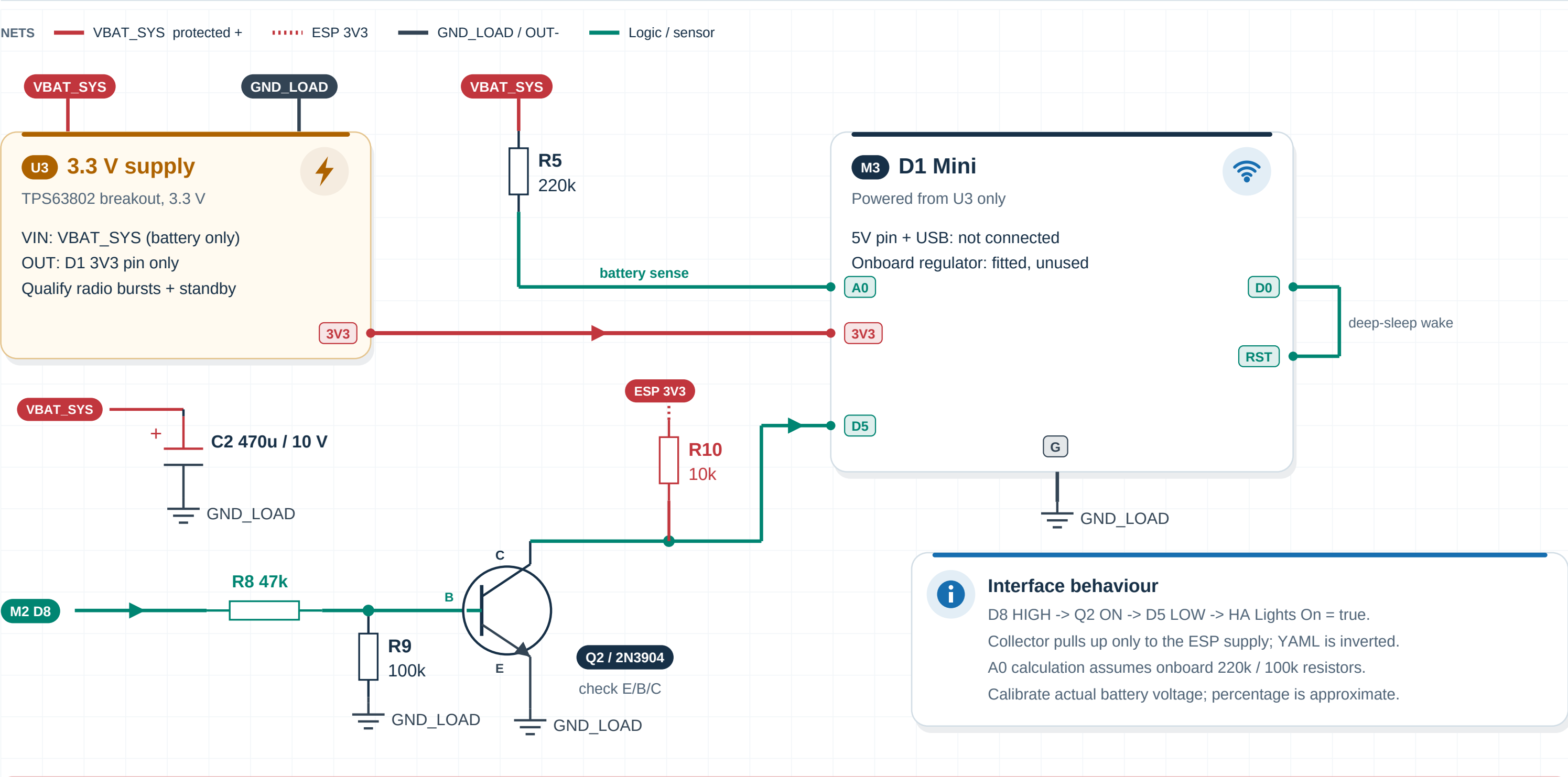
Light controller & LED driver

Reused 16 MHz Pro Mini converted to internal 8 MHz. Switched LDR; dusk to dawn at 100% command, with R3 setting the current.



Low-power telemetry reference (D1 Mini)

Retained for the later production profile. This is not the always-on D1 Mini UART test system on sheet 6.



SAFETY Disconnect external power AND signal leads before USB/programmer servicing. Never supply the ESP 3V3 pin from the cell directly.

WEATHERPROOF BOX / SHADED POSITION

PANEL

LIGHTS

LDR

PERFBOARD / MODULE CARRIER

U0 Input 5 V
Open item - not fitted

M1 Charger
IN / B / OUT labelled

M2 Pro Mini
8 MHz; sockets; ISP access

LED driver
R3 only - D9 direct

Protected rails

U3 3V3
TPS63802

M3 D1
Sockets / service

ANTENNA

Keep antenna clear

BT1 / 18650

BT2 / 18650

Insulated parallel holder; fuse each + lead; separate cells from heat.

Build in this order

- 1 Qualify panel regulator + charger.
- 2 Convert and test the Pro Mini.
- 3 Add R3 (47R) and light string - no MOSFET.
- 4 Add LDR; check dusk/dawn + LVC.
- 5 Assemble the protected, fused pack.
- 6 Add optional telemetry and measure.

Label every connector

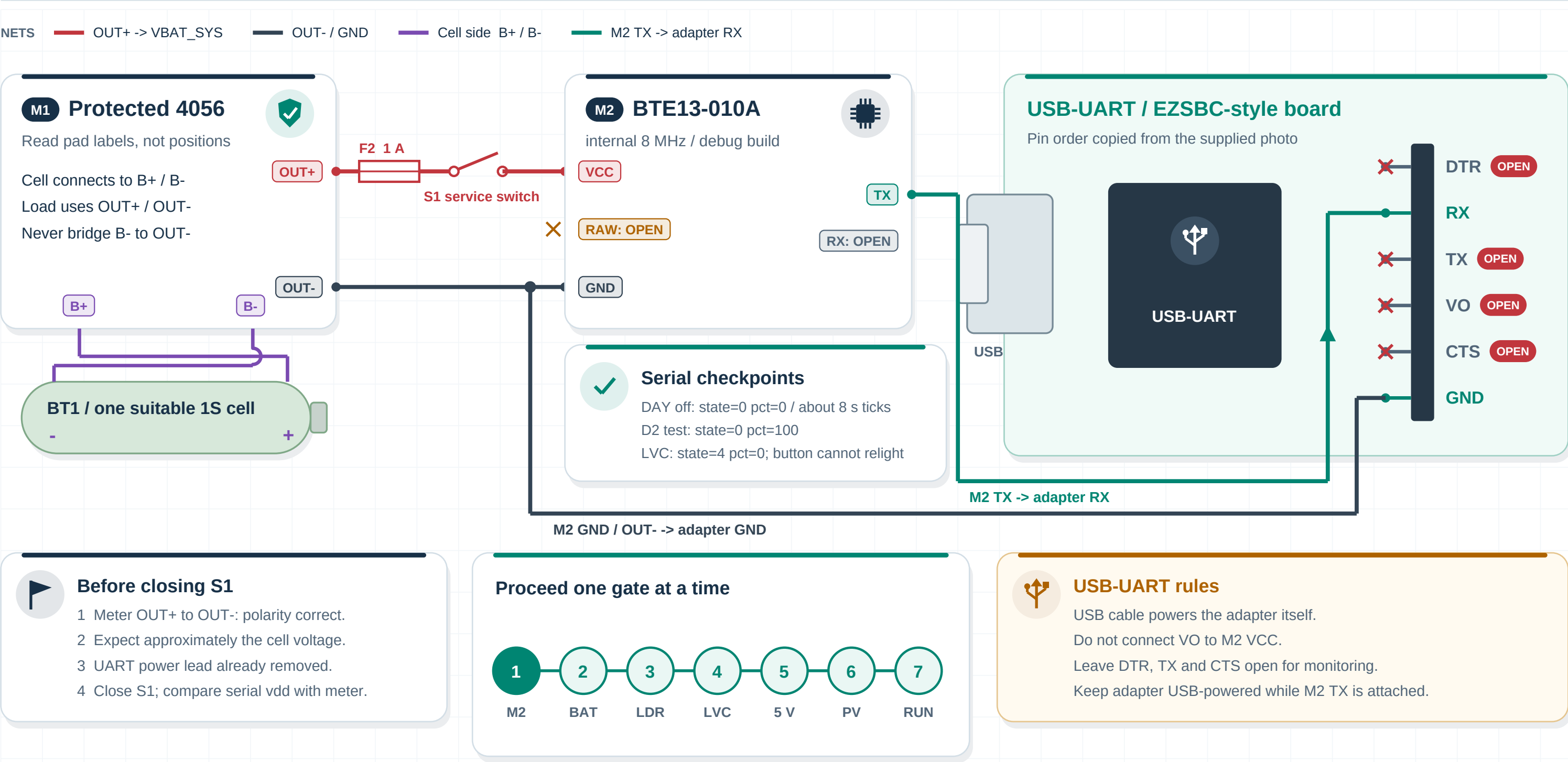
- RED: protected positive / VBAT_SYS
- BLACK: load return / OUT-
- BLUE: panel input pair
- PURPLE: battery-only connections
- GREEN: logic / sensor signals

Before closing the lid

- ☐ Strain relief, polarity, cell insulation.
- ☐ No exposed underside copper.
- ☐ Charger/cell temperature tests passed.
- ☐ Log charge, load and actual brightness.

Staged power-up & serial diagnostics

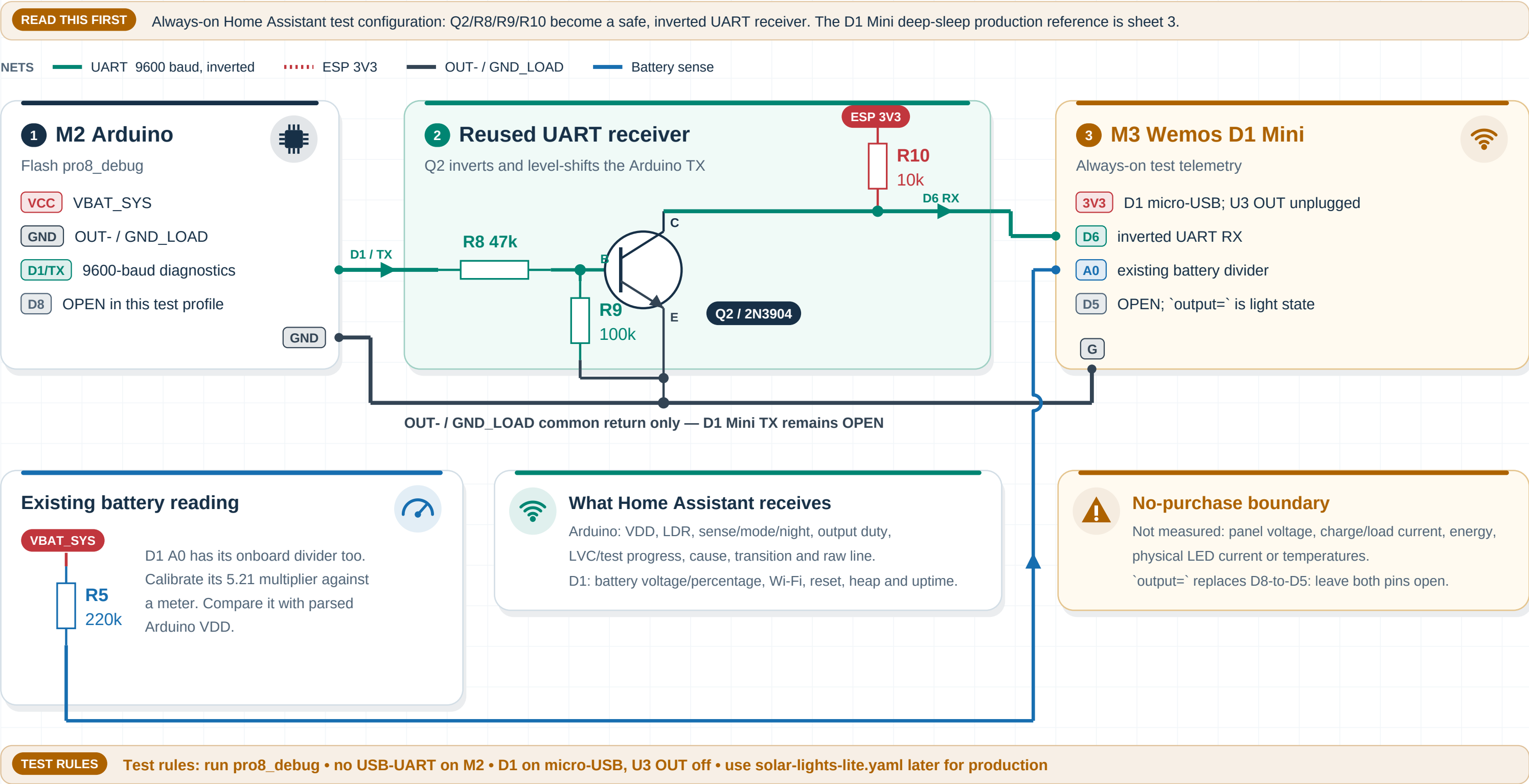
Battery powers the controller. The USB-UART listens only: connect RX and GND; leave its power and control pins open.



RULE SERIAL IS NOT A CHARGE METER: verify panel input, regulated 5 V, battery voltage and charge current with instruments.

Development telemetry (Wemos D1 Mini)

TEST SYSTEM ONLY: always-on Wi-Fi and UART capture. Reuses the existing status transistor parts; no INA226 or new telemetry module. Production D1 profile remains on sheet 3.



Current architecture schematic

As-built v0.5 wiring. The development receiver captures the Arduino UART in Home Assistant; it is not a current or energy meter.

